

Hill Choy

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SUMMARY

MSc student at University College London with a BEng in Computer Engineering from HKUST. Experienced in developing and optimizing Computer Vision models for real-time Edge AI applications, including traffic and pedestrian monitoring systems. Focused on the practical implementation of machine learning to solve real-world visual computing challenges.

EDUCATION

University College London (UCL)

MSc Computer Graphics, Vision and Imaging

Expected Graduation: September 2026

- Highlighted Coursework:

- Highlighted Coursework: Machine Vision, ML for Visual Computing, Inverse Problems, Robot Vision.

Hong Kong University of Science and Technology (HKUST)

BEng in Computer Engineering (Second Class Honors, Division 1)

Graduation: July 2024

- Highlighted Coursework:

- Highlighted Coursework: Intro to NLP, Computer & Communication Networks.

IELTS Academic

8 (Out of 9), (R:8.5,L:9,W:7,S:7)

RESEARCH EXPERIENCE

Hyperspectral Image Analysis for Safe-Fabric Segmentation

UCL, MSc Industry Dissertation Project (Partner: Zori Tex) (Ongoing)

- Collaborating with industry partner Zori Tex to develop an advanced computer vision pipeline for analyzing complex textile imagery.

- Designing robust segmentation and classification algorithms to generate precise safe-fabric masks and categorize colour families. The pipeline is engineered to extract reliable features while actively mitigating the challenges of high-dimensional hyperspectral data and uneven real-world illumination.

Advanced Video Analytics and Edge AI for Smart Carpark Systems

HKUST, Bachelor Degree Final Year Project (Supervisor: Prof. Shueng-Han Gary Chan, CSE, HKUST)

- Developed a smart car park system that utilizes edge AI to provide real-time parking space availability while minimizing computational, installation, and maintenance costs.

- Developed a YOLOv8 model on a custom low-light/wide-angle dataset for Raspberry Pi. Enabled low-power deployment via mobile power banks by integrating a MobileNetV3 backbone to prune parameters and converting to ONNX format for low-precision inference. This pipeline achieved highly efficient, low-precision inference while preserving robust detection accuracy.

PROFESSIONAL EXPERIENCE

Research Assistant

Prof. Shueng-Han Gary Chan, CSE, HKUST

Hong Kong

08/2024 - 06/2025

- List of Projects:

- K11 MUSEA Cars and Pedestrian Flow Monitoring System
 - Developed an end-to-end traffic monitoring system for a mall complex, including the YOLO model training, data capture pipelines, and Flask-based web frontend.
- Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera
 - Developed a privacy-conserving monitoring system for elderly care that utilizes edge AI to detect falls and issue real-time alerts to staff. Designed and implemented an online web portal featuring automated data visualization and daily report generation scripts to streamline patient health tracking.
 - Recipient of the **Hong Kong ICT Awards 2024, Smart People (Smart Ageing) Silver Award**.
- Smart Traffic Management System (Kung Tong, Hong Kong)
 - Developed real-time traffic information capture pipelines to monitor major junctions, enabling pattern analytics and automated alerts for congestion or accidents to local authorities. Responsible for the end-to-end design of data pipelines to support large-scale urban traffic management.
 - Link: <https://www.police.gov.hk/>

TECHNICAL SKILLS & LANGUAGES

- **Technical Stack:** Python, C++, Flask, PyTorch, YOLOv8, OpenCV, Git, Raspberry Pi, NVIDIA Jetson Nano
- **Languages:** Cantonese(Native), Mandarin(Basic), English(Fluent), Japanese(Foundation)